
Auditing Feature Importance Rankings in Nonstationary Data Streams

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Abstract: Feature importance explanations are often evaluated on fixed tabular data, although many deployed models operate on nonstationary streams in which the data distribution and relevant signal may change over time. In such settings, a ranking can appear stable while failing to respond to drift, or change for reasons unrelated to the label signal learned by the model. This paper proposes a protocol for evaluating feature importance rankings in data streams. The protocol combines a sanity gate based on label randomization, deletion tests for faithfulness, keep-top- k sufficiency, temporal rank stability, negative controls, and Drift-Explanation Alignment (DEA), a diagnostic that relates distributional drift to changes in feature rankings. We evaluate the protocol on controlled synthetic streams representing stable, abrupt, gradual, and correlated proxy conditions, as well as selected real datasets from electricity markets, insect monitoring, gas sensing, and industrial fault detection. Controls that are independent of the trained model fail the sanity gate, even when they are perfectly stable or responsive to drift. Rankings derived from trained models recover known signal features in controlled streams but show mixed reliability on real data. These findings support interpreting DEA and temporal stability only after establishing model dependence.

Keywords: Explainable AI, data streams, concept drift, feature importance, explanation evaluation, sanity checks, trustworthy AI

1. Introduction

Feature importance rankings are widely used to summarize model behaviour in structured data applications. Methods such as LIME and SHAP have helped establish post hoc feature attribution as a practical way to explain complex predictions [1, 2]. We use LIME and SHAP as examples from the broader attribution literature, while our experiments focus on lightweight rankings computed globally or within individual chunks. These include linear coefficients, permutation importance, and native tree importance, which support reproducible evaluation in streaming settings. Feature rankings are easy to compute and present, and they are commonly used for debugging, auditing, and communicating model behaviour. Their apparent simplicity is also a weakness. A ranking can appear plausible while remaining insensitive to the learned model, stable for the wrong reason, or dependent on arbitrary perturbation choices. Previous sanity and robustness studies have shown that explanations may remain similar after model parameters or labels are randomized, or may change under transformations that should not affect predictions [3–5]. These findings motivate conservative evaluation based on multiple complementary diagnostics rather than reliance on a single explanation score.

Most protocols for evaluating feature importance implicitly assume a static setting in which the model, data distribution, and interpretation target are assessed on a fixed dataset. This assumption is restrictive for applications involving data streams. In nonstationary environments, the input distribution may shift, the features carrying predictive signal may change, and the model may be retrained or updated repeatedly [6, 7].

Under these conditions, explanation reliability has a temporal dimension. A reliable ranking should not merely be reproducible across random seeds. It should remain stable while the underlying concept is stable and change when the relevant signal changes. Conversely, a ranking that changes whenever marginal feature variance changes may appear responsive to drift while remaining disconnected from the learned decision function.

This paper studies the evaluation of feature importance under controlled forms of drift in data streams. We focus on global rankings and rankings computed within individual chunks for structured data classification. Our aim is not to introduce a new explanation method, but to define an auditable workflow for determining whether explanations remain dependent on the trained model, faithful, sufficient, temporally stable, and responsive to drift. The paper makes four contributions:

1. We propose a sanity-gated evaluation protocol for feature importance rankings that explicitly accounts for drift in nonstationary data streams.
2. We define and evaluate Drift–Explanation Alignment (DEA), a diagnostic that relates distributional drift to temporal changes in feature importance rankings.
3. We evaluate rankings obtained from trained models alongside controls that are independent of the model, using controlled streams and selected real datasets evaluated as ordered streams.
4. We show that temporal stability and responsiveness to drift are not sufficient evidence of reliability unless they are interpreted alongside sanity checks, faithfulness, sufficiency, and the behaviour of negative controls.

2. Related Work

Explanation sanity and robustness. Sanity checks examine whether explanations are sensitive to the learned model and its label signal. Adebayo et al. [3] showed that some saliency explanations remain visually similar after model randomization, while Kindermans et al. [4] and Ghorbani et al. [5] identified problems involving invariance and fragility. Although these studies were largely motivated by image explanations, their central lesson also applies to structured data: plausibility and reproducibility alone do not establish model dependence. A ranking that remains similar under true and randomized labels should therefore not be accepted as evidence of model dependence without further checks.

Faithfulness, sufficiency, and stability. Perturbation and retraining approaches, including deletion, insertion, ROAR, and ROAD, examine whether features identified as important affect model behaviour [8–10]. Deletion measures whether performance deteriorates when the most important features are removed, whereas sufficiency measures whether retaining only those features preserves performance. These diagnostics can disagree, particularly when features are redundant. Stability assesses whether rankings remain consistent across runs or perturbations [11], but a fixed ranking can be perfectly stable while remaining unfaithful. Toolkits such as Quantus and OpenXAI support systematic explanation evaluation [12,13]. However, reporting multiple metrics does not by itself determine how conflicting outcomes should be interpreted.

Data streams and concept drift. Research on concept drift studies learning when data distributions or decision boundaries change over time [6,7]. Most of this literature concentrates on drift detection and model adaptation rather than explanation reliability. Recent studies have begun to address explanations in evolving environments. Muschalik et al. [23] use feature importance to explain changes between models, Haug et al. [24] detect changes in local explanations over evolving streams, and Fumagalli et al. [25] introduce incremental permutation feature importance for producing global explanations under concept drift. Complementary work examines feature level drift [19], explanatory monitoring of performance under feature shifts [21], and the broader relationship between explainability and drift [20,22]. These studies show how explanations can be generated, compared, or monitored in dynamic settings. They do not, however, provide the combined audit considered here, in which model dependence is tested before faithfulness, sufficiency, temporal behaviour, and drift response are interpreted alongside negative controls. This motivates a dedicated evaluation protocol for feature importance rankings in nonstationary streams.

3. Drift-Aware Sanity-Gated Protocol

Consider a stream divided into chunks $C_1, \dots, C_T$, where chunk C_t contains structured features $X_t \in \mathbb{R}^{n_t \times d}$ and labels y_t available for retrospective evaluation. A classifier f_t is trained on a fit partition from the current chunk, and an explanation method produces a non-negative feature importance vector $s_t \in \mathbb{R}^d$. The protocol evaluates each chunk separately and then summarizes explanation behaviour across the stream.

3.1. Chunk-wise sanity gate

For each ranking obtained from a trained model, we retrain the same model after randomly permuting the fit labels within the chunk and then recompute its feature importance vector. Let s_t denote the vector obtained with the true labels, and let $s_{t,\ell}^{\text{shuf}}$ denote the vector from label randomization repetition ℓ . Define $\mathcal{V}_t$ as the set of repetitions for which the Spearman correlation is defined. The reported gate statistic is

$$\bar{\rho}_t^{\text{gate}} = \frac{1}{|\mathcal{V}_t|} \sum_{\ell \in \mathcal{V}_t} |\rho_s(s_t, s_{t,\ell}^{\text{shuf}})|, \quad (3.1)$$

provided that $\mathcal{V}_t$ is nonempty. When all L comparisons are defined, the outcome is *pass* if $\bar{\rho}_t^{\text{gate}} < \tau_p$, *inconclusive* if $\tau_p \leq \bar{\rho}_t^{\text{gate}} < \tau_f$, and *fail* otherwise. We use $\tau_p = 0.20$ and $\tau_f = 0.30$ as fixed diagnostic bands and also examine stricter $(0.15, 0.25)$ and more lenient $(0.25, 0.35)$ settings. A formal alternative would estimate a null distribution tailored to the method, dimensionality, and chunk using many label randomizations; the present lightweight design does not support that inferential interpretation.

Two equal constant vectors receive correlation one. Any other comparison involving a constant vector contains no rank information and is undefined. If at least one of the L comparisons is undefined, the chunk is conservatively marked inconclusive, although the mean over defined comparisons is retained for descriptive reporting. The controls FIXEDRANK, RANDOMRANK, and VARIANCERANK do not use labels, so their rankings remain unchanged after label randomization and fail the gate by design.

3.2. Faithfulness, sufficiency, and conflict

Let S_t^k contain the top k features according to s_t . The deletion test replaces the features in S_t^k with their means estimated from the fit partition, which are zero after standardization, and measures the resulting decrease in test accuracy. The sufficiency test retains the features in S_t^k , masks their complement using the same values, and measures the resulting test accuracy. Both results are compared with matched random subsets containing the same number of features.

Deletion evidence is considered positive when the observed accuracy drop exceeds the mean drop from the random subsets by more than half of their sample standard deviation. Sufficiency evidence is positive when the accuracy obtained by retaining the top k features exceeds the corresponding random subset mean by the same margin. Each chunk that does not fail the sanity gate is assigned one of four exploratory outcomes: consistent evidence from both tests, influence only, sufficiency only, or weak or absent evidence. Chunks that fail the gate are excluded from this interpretation. Inconclusive chunks are retained for exploratory analysis but are not counted as gate passes.

3.3. Temporal stability and Drift–Explanation Alignment

For $t = 2, \dots, T$, temporal stability is measured by the Spearman correlation between consecutive feature importance vectors. Rank shift is defined as

$$\Delta R_t = \frac{1 - \rho_s(s_t, s_{t-1})}{2}, \quad (3.2)$$

which maps Spearman correlation from $[-1, 1]$ to a rank shift scale in $[0, 1]$. Larger values therefore indicate greater change in the ranking. Undefined temporal correlations are omitted from temporal summaries.

Let W_1 denote the one-dimensional Wasserstein distance. The raw distributional drift between consecutive chunks is

$$W_t = \frac{1}{d} \sum_{j=1}^d W_1 \left(X_{t-1}^{(j)}, X_t^{(j)} \right), \quad D_t = \frac{W_t}{\max_{2 \leq u \leq T} W_u}, \quad (3.3)$$

where normalization is performed retrospectively within each seed and stream. We set $D_1 = 0$ and $\Delta R_1 = 0$ by convention. This normalization produces $D_t \in [0, 1]$ for the evaluated sequence. A prospective deployment would instead require a fixed reference scale or a scale updated using past observations only.

Drift–Explanation Alignment is defined as

$$\text{DEA}_t = D_t \Delta R_t. \quad (3.4)$$

DEA is a co-occurrence diagnostic rather than a reliability score or causal measure. It is high only when distributional drift and ranking change occur together and should therefore be reported with both D_t and ΔR_t . After a method passes the sanity gate, a localized increase in DEA near an established change point is consistent with explanation adaptation. Persistently high DEA, high DEA without perturbation evidence, or substantial rank shift when D_t is low instead suggests instability or noise. Conversely, low DEA is appropriate during a stable period but may be concerning when a known signal shift should alter the ranking. We therefore use DEA for diagnostic triage and interpret it only after model dependence, faithfulness, and sufficiency have been examined.

4. Experimental Design

4.1. Controlled synthetic stream settings

The primary evaluation uses four controlled synthetic streams for which the signal-generating feature groups are known. Each stream contains 16 features, eight chunks of 300 observations, and three random seeds. This gives 96 chunk evaluations across the four stream settings for each model and ranking combination. Binary labels are generated from noisy logits and balanced within each chunk using the median logit as the decision threshold.

Table 1. Controlled synthetic streams and their diagnostic roles.

Stream	Signal movement	Diagnostic purpose	Expected explanation behaviour
Stable control	Active group fixed across chunks	Tests false temporal variation	Variation should reflect sampling noise only
Abrupt signal shift	Active group changes at midpoint	Tests sudden response to signal change	Rank shift should increase near the transition
Gradual signal shift	Relevance transfers smoothly	Tests progressive explanation adaptation	Ranking change should occur progressively
Correlated proxy shift	Signal moves through correlated features	Tests redundancy and proxy effects	Deletion and sufficiency may disagree

Table 1 defines the expected qualitative behaviour of each controlled stream. These expectations guide interpretation but are not used as pass or fail criteria.

4.2. Selected real-data stress checks

We additionally evaluate the protocol on four benchmark datasets treated as ordered streams. ELEC contains electricity market observations and provides a compact example of temporal distribution change [14, 15]. INSECTS-abrupt is derived from insect sensor measurements ordered to produce abrupt changes across operating conditions [16]. Gas Sensor Array contains multivariate chemical sensor responses [17], while DAMADICS provides measurements associated with industrial actuator operation and faults [18]. Because these datasets do not provide chunk-level ground-truth feature rankings, they are used to assess model dependence, perturbation behaviour, temporal stability, and drift alignment rather than exact signal recovery.

For computational consistency, Gas Sensor Array is restricted to the 48 features with the highest global variance. This is a retrospective preprocessing choice and is considered when interpreting the variance-based control.

Table 2. Benchmark datasets used in the real-data stress checks. Rows and features are reported after preprocessing, and chunks are counted per seed.

Dataset	Rows	Features used	Classes	Chunks	Role in evaluation
ELEC	45,312	8	2	10	Temporal electricity market drift
INSECTS-abrupt	52,848	33	6	10	Abrupt sensor drift benchmark
GasSensorArray	13,910	48/128	6	10	High-dimensional sensor stress check
DAMADICS-binary	3,600	32	2	8	Industrial normal/fault monitoring

Table 2 reports the dimensions after preprocessing and the number of chunks evaluated for each dataset. For GasSensorArray, 48/128 indicates that the 48 numerical features with the highest global variance were retained from the 128 available features.

4.3. Models, rankings, and controls

The controlled experiments use logistic regression and a depth-limited decision tree. The real-data experiments use a linear classifier trained with log loss and a separately configured depth-limited decision tree. Class weighting is used to reduce sensitivity to imbalance.

For linear models, features are ranked by the absolute coefficient magnitude; for multiclass models, the absolute coefficients are averaged across classes. Tree models use native impurity-based importance. Permutation importance is also evaluated using validation accuracy as the scoring measure.

Three analytical controls are included. FIXEDRANK uses the original column order. RANDOMRANK uses a seed-specific random order that remains fixed across chunks. VARIANCERANK ranks features by their marginal variance in the fit partition of each chunk. The first two controls test whether the protocol rejects rankings that are temporally stable but unrelated to the model or labels. VARIANCERANK tests whether responsiveness to marginal distribution change can be mistaken for model-dependent explanation.

4.4. Implementation details

Within each chunk, 30% of the observations are reserved for testing. The remaining observations are divided into fit and validation partitions, with 30% of that remainder used for validation. Standardization parameters are estimated from the fit partition only. Feature rankings are computed on the validation partition, while deletion and sufficiency diagnostics are evaluated on the held-out test partition.

The number of retained or deleted features is

$$k = \max(1, \text{round}(0.25d)),$$

where d is the number of input features. Masked standardized features are set to zero, which corresponds to the fit-partition mean.

The experiments use three random seeds, two label-randomization repetitions, five matched Random- k subsets, and one permutation-importance repetition. Real-data chunks are capped at 1,500 observations for computational consistency. Missing numerical values are median imputed using the supplied dataset before chunking. Software versions, random seeds, model settings, dataset sources, and preprocessing choices are provided with the accompanying reproducibility materials.

5. Results

5.1. Controlled streams: sanity, drift response, and signal recovery

Across the controlled streams, all analytical controls fail the gate in every chunk. FIXEDRANK and RANDOMRANK remain perfectly stable, whereas VARIANCERANK responds to changing marginal variances; neither property establishes model dependence. Model-based rankings show mixed outcomes: logistic-regression permutation and tree-native importance pass in 59 and 56 of 96 chunks, compared with 38 for coefficients and 39 for tree permutation. The 44 inconclusive tree-permutation outcomes, including three undefined shuffled-label comparisons, warrant particular caution.

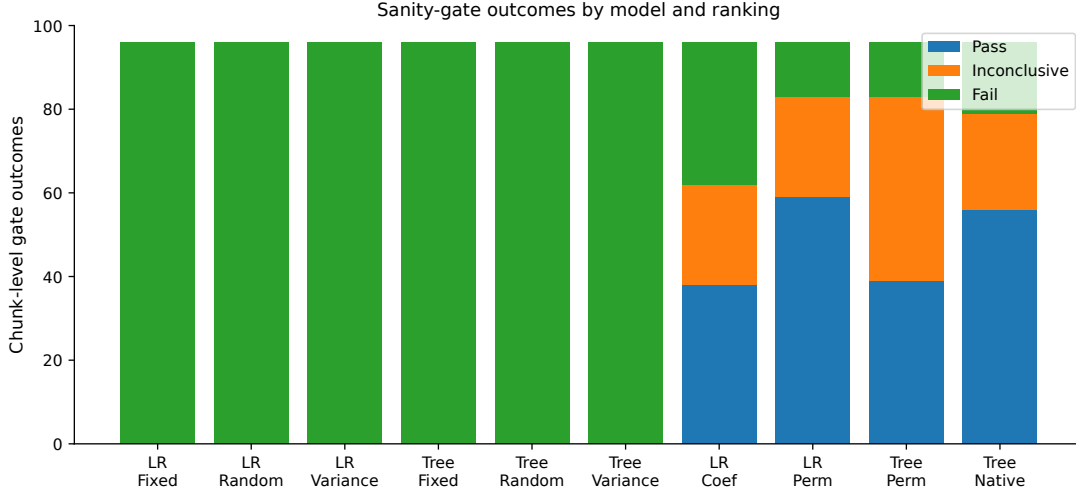


Figure 1. Controlled-stream sanitary-gate outcomes by model and ranking. Analytical negative controls fail by design in every chunk, while model-based rankings show mixed pass, inconclusive, and fail outcomes.

Table 3. Controlled-stream reliability summary. Gate counts are chunk-level outcomes. DEA is Drift–Explanation Alignment; overlap is mean top- k overlap with the known active signal group.

Model	Ranking	Type	Pass	Inc.	Fail	$\bar{\rho}^{\text{gate}}$	Stability	DEA	Overlap
LogReg	FIXEDRANK	control	0	0	96	1.000	1.000	0.000	0.656
LogReg	RANDOMRANK	control	0	0	96	1.000	1.000	0.000	0.305
LogReg	VARIANCERANK	control	0	0	96	1.000	0.033	0.259	0.357
Tree	FIXEDRANK	control	0	0	96	1.000	1.000	0.000	0.656
Tree	RANDOMRANK	control	0	0	96	1.000	1.000	0.000	0.305
Tree	VARIANCERANK	control	0	0	96	1.000	0.033	0.259	0.357
LogReg	Coef	model	38	24	34	0.249	0.566	0.130	0.995
LogReg	Perm	model	59	24	13	0.194	0.617	0.114	0.964
Tree	Perm	model	39	44	13	0.218	0.454	0.146	0.789
Tree	TREEIMP	model	56	23	17	0.188	0.450	0.150	0.883

Known-signal overlap is substantially higher for model-based rankings than for controls: 0.995 for coefficients, 0.964 for logistic-regression permutation, 0.883 for tree importance, and 0.789 for tree permutation. Coefficients reach 1.000 in three streams and 0.979 under gradual drift, as expected from the generators’ linear signal structure. This is a positive signal-recovery check, not evidence that coefficients are generally sufficient under arbitrary drift.

Figure 1 visualizes the gate outcomes. Figure 2 relates normalized drift to rank shift: model-based rankings generally change more in high-drift regions, but VARIANCERANK also changes despite failing the gate. Mean model-based DEA rises from 0.031 to 0.163 across the abrupt shift and from 0.035 to 0.212 across the correlated-proxy shift. These initial-phase means include the zero-valued first chunk by convention; excluding it changes the values to 0.042 and 0.047, respectively, without changing the phase comparison.

Table 4 shows modest but nonzero deletion–sufficiency conflict, confirming that influential and sufficient feature sets need not coincide. Figure 3 shows how overlap for model-based rankings evolves across chunks, while Table 3 shows that their aggregate overlap exceeds that of the controls.

Table 4. Controlled-stream deletion–sufficiency outcomes for chunks that pass or are inconclusive. Inconclusive chunks remain exploratory and are not counted as passes.

Model	Ranking	Eligible n	Both	Influence only	Sufficiency only	Weak/no evidence	Conflict rate
LogReg	Coef	62	58	1	3	0	0.065
LogReg	Perm	83	77	4	2	0	0.072
Tree	Perm	83	75	1	3	4	0.048
Tree	TREEIMP	79	76	0	3	0	0.038

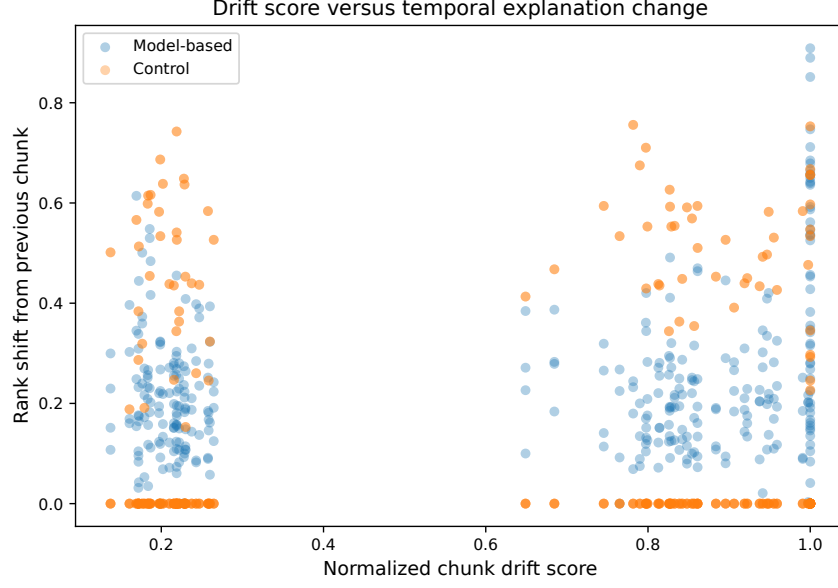


Figure 2. Relationship between normalized distributional drift and temporal rank shift. DEA captures their product, but the sanity gate is required to distinguish model-dependent explanation change from control behaviour.

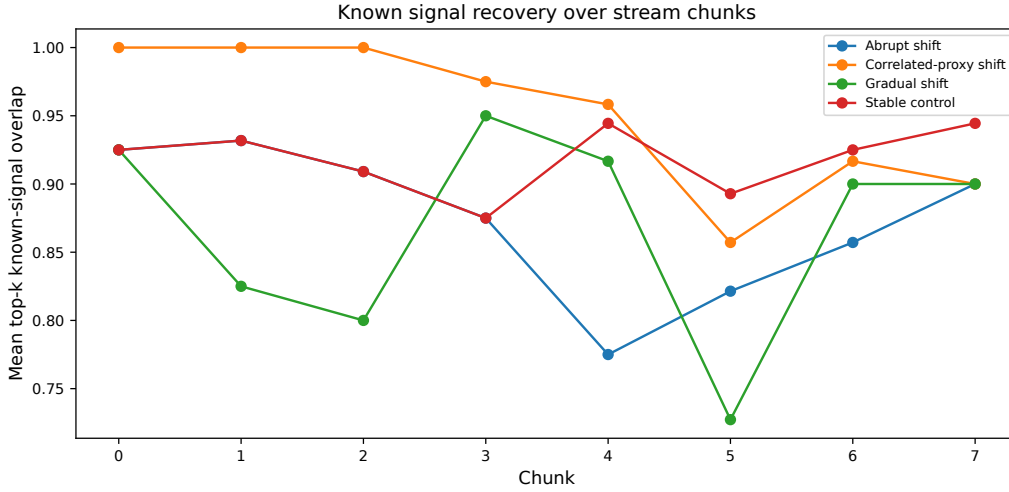


Figure 3. Mean top- k overlap with known active signal features by stream and chunk, averaged over model-based rankings, seeds, and chunks that do not fail the gate.

5.2. Selected real-stream stress check

Table 5 reports the real-data check. All 456 control evaluations fail: fixed and random rankings remain stable, while the variance control has nonzero DEA but remains label independent. Model-based reliability is mixed. Coefficients have the weakest gate profile (33/11/32) and highest conflict rate (0.318). Five permutation comparisons without rank information are inconclusive. Tree permutation has no observed conflict among eligible chunks and tree-native importance has a rate of 0.016, but these descriptive results do not establish superiority. ELEC has the least favourable gate profile; the experiment does not isolate why. The GasSensorArray variance cap may favour variance rankings, yet VARIANCERANK still fails every gate.

Stricter thresholds (0.15,0.25) yield controlled pass/inconclusive/fail counts of 124/130/130; lenient thresholds (0.25,0.35) yield 251/86/47. Real-data counts are 135/82/87 and 212/32/60. Undefined cases

Table 5. Selected real-stream stress check. Gate entries report pass/inconclusive/fail counts over two seeds. Controls are summarized across both models and all streams. Conflict uses model-based chunks that do not fail.

Model	Ranking	Gate P/I/F	$\bar{p}^{\text{gate}}$	DEA	Conflict
Linear	Coef	33/11/32	0.307	0.178	0.318
Linear	Perm	47/15/14	0.211	0.171	0.177
Tree	Perm	46/18/12	0.193	0.194	0.000
Tree	TREEIMP	51/12/13	0.209	0.190	0.016
Controls	FIXEDRANK	0/0/152	1.000	0.000	—
Controls	RANDOMRANK	0/0/152	1.000	0.000	—
Controls	VARIANCERANK	0/0/152	1.000	0.038	—

remain inconclusive and all controls fail, confirming that these thresholds are diagnostic settings rather than universal labels.

6. Discussion

Three lessons emerge. Model dependence should gate interpretation because stability or drift response can occur without label dependence. Temporal stability must be judged relative to drift, and deletion and sufficiency remain distinct, particularly with correlated proxies. Controlled streams establish these points with known signal; real-data checks test them without feature-level ground truth.

DEA is determined by normalized drift and rank shift, so it adds no independent information. It is a co-occurrence summary for triage, not evidence of causal adaptation or a standalone score. Because normalization is within each seed–stream pair, DEA is sequence relative and cross-stream averages are descriptive.

Three controlled seeds, two real-data seeds, two label shuffles, and five Random- k subsets support descriptive rather than inferential conclusions. The half-standard-deviation rule is heuristic; larger studies should increase repetitions, report uncertainty, and use empirical nulls. Mean masking can break dependencies and create out-of-distribution combinations, motivating distribution-preserving perturbations.

Real-data preprocessing is retrospective: global variance filtering and median imputation use the full supplied table. Deployment must estimate these parameters from past data. LIME, SHAP, adaptive online learners, and delayed-label settings remain future work.

7. Conclusion

We proposed a sanity-gated audit of feature rankings under drift. It rejects label-independent rankings despite apparent stability or drift response, while model-based results remain mixed on benchmarks. DEA is a co-occurrence diagnostic, not a standalone reliability score.

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